UNIVERSITÀ DEGLI STUDI DI PISA
CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta del 27/06/2024*

EXERCISE

We observe N samples of a realization of the random process $X[n] = A + W[n]$, for n=1,2,…, N. where A a constant signal of interest (SOI), modelled as an unknown deterministic parameter, and $\{W[n]; n = 1, 2, \cdots, N\}$ are independent and identically distributed (IID) random variables modeling additive white non-Gaussian noise. Each noise sample is uniformly distributed between -1 and 1.

(1) Derive the signal to noise ratio (SNR) $\gamma \triangleq A^2 \big/ E\{W^2[n]\}$, the autocorrelation function (ACF) $R_X[m] = E\{X[n]X[n+m]\}$, and the power spectral density (PSD) $S_X\left(e^{j2\pi f}\right) = FT\{R_X[m]\}$ of the observed signal $X[n]$. Plot $R_X[m]$ and $S_X\left(e^{j2\pi f}\right)$.

(2) Derive the Method of Moments (MM) estimator of the unknown parameter A.

(3) Derive the bias and the Mean Square Error (MSE) of $\hat{A}_{MM}$. Plot the MSE as a function of N and find the minimum value of N that guarantees an MSE lower than 10^{-3}. Discuss the consistency of $\hat{A}_{MM}$.

(4) Derive the Maximum Likelihood (ML) estimator of the unknown parameter A when N=1, N=2 and N is generic.

Solution:

(1) The signal to noise ratio $\gamma \triangleq A^2 / E\{W^2[n]\}$

$$\sigma_W^2 = E\{W^2[n]\} = \int_{-\infty}^{+\infty} w^2 f_W(w)dw = \frac{1}{2}\int_{-1}^{+1} w^2 dw = \int_{0}^{+1} w^2 dw = \frac{1}{3}$$

$$\gamma = A^2 / E\{W^2[n]\} = 3A^2$$

The autocorrelation function (ACF) and the power spectral density (PSD) of the signal $X[n]$ are:

$$\begin{aligned} R_X[m] &= E\{X[n]X[n+m]\} = E\{(A+W[n])(A+W[n+m])\} = A^2 + E\{W[n]W[n+m]\} \\ &= A^2 + \sigma_W^2 \delta[m] = A^2 + \frac{1}{3}\delta[m] \end{aligned}$$

$$S_X\left(e^{j2\pi f}\right) = FT\{R_X[m]\} = 2\pi A^2 \delta\left(e^{j2\pi f} - 1\right) + \sigma_W^2 = 2\pi A^2 \delta\left(e^{j2\pi f} - 1\right) + \frac{1}{3}$$

(2) Method of Moments (MM) estimator of the unknown parameter A.

$$m_1 = E\{X[n]\} = A \quad \Rightarrow \quad \hat{A}_{MM} = \hat{m}_1 = \frac{1}{N}\sum_{n=1}^{N} X[n]$$

(3) Bias and the mean square error (MSE) of the MM estimator of A:

$$b\{\hat{A}_{MM}\} = E\{A - \hat{A}_{MM}\} = A - E\left\{\frac{1}{N}\sum_{n=1}^{N} X[n]\right\} = A - \frac{1}{N}\sum_{n=1}^{N} A = 0 \quad \Rightarrow \quad \text{unbiased}$$

$$MSE\{\hat{A}_{MM}\} = \text{var}\{\hat{A}_{MM}\} = \frac{\text{var}\{X[n]\}}{N} = \frac{\text{var}\{W[n]\}}{N} = \frac{\sigma_W^2}{N} = \frac{1}{3N}$$

$\hat{A}_{MM}$ is **consistent**, since its MSE tends to 0 when N tends to infinity.

$$MSE\{\hat{A}_{MM}\} = \frac{1}{3N} < 10^{-3} \quad \rightarrow \quad N > \frac{10^3}{3} = 334$$

(4) Maximum Likelihood (ML) estimator of the unknown parameter A when N=1, N=2 and N is generic.

Let's define $x_i \triangleq X[i], \quad i = 1, 2, \cdots, N$

<u>N=1</u>:

$$L(A) = f_X(x_1; A) = \frac{1}{2} rect\left(\frac{x_1 - A}{2}\right)$$

$$\hat{A}_{ML} = \arg\max_A L(A) = \arg\max_A rect\left(\frac{x_1 - A}{2}\right) = \{\text{any value from } x_1 - 1 \text{ to } x_1 + 1\}$$

Since $L(A) = \begin{cases} 1, & x_1 - 1 < A < x_1 + 1 \\ 0, & otherwise \end{cases}$

If we want to select the one that is *unbiased* we select the middle-point of the interval:

$$\hat{A}_{ML} = \frac{(x_1 - 1) + (x_1 + 1)}{2} = x_1 \quad \Rightarrow \quad E\left\{\hat{A}_{ML}\right\} = E\{x_1\} = A$$

<u>N=2</u>:

$$L(A) = f_X(\mathbf{x}; A) = f_X(x_1; A) f_X(x_2; A) = \frac{1}{4} rect\left(\frac{x_1 - A}{2}\right) rect\left(\frac{x_2 - A}{2}\right)$$

$$\hat{A}_{ML} = \arg\max_A L(A) = \arg\max_A rect\left(\frac{x_1 - A}{2}\right) rect\left(\frac{x_2 - A}{2}\right)$$

$$= \{\text{any value from } Max\{x_1, x_2\} - 1 \text{ to } \min\{x_1, x_2\} + 1\}$$

Since $L(A) = \begin{cases} 1, & Max\{x_1, x_2\} - 1 < A < \min\{x_1, x_2\} + 1 \\ 0, & otherwise \end{cases}$

If we want to select the one that is *unbiased* we select the middle-point of the interval:

$$\hat{A}_{ML} = \frac{\left(Max\{x_1, x_2\} - 1\right) + \left(\min\{x_1, x_2\} + 1\right)}{2} = \frac{\min\{x_1, x_2\} + Max\{x_1, x_2\}}{2} = \frac{x_1 + x_2}{2}$$

$$E\left\{\hat{A}_{ML}\right\} = E\left\{\frac{x_1 + x_2}{2}\right\} = \frac{E\{x_1\} + E\{x_2\}}{2} = A$$

<u>*N* generic</u>:

$$L(A) = f_X(\mathbf{x}; A) = \prod_{i=1}^{N} f_X(x_i; A) = \prod_{i=1}^{N} \frac{1}{2} rect\left(\frac{x_i - A}{2}\right) = \frac{1}{2^N} \prod_{i=1}^{N} rect\left(\frac{x_i - A}{2}\right)$$

$$\hat{A}_{ML} = \arg\max_A L(A) = \arg\max_A \prod_{i=1}^{N} rect\left(\frac{x_i - A}{2}\right) = \left\{\text{any value from } Max\{x_i\} - 1 \text{ to } \min\{x_i\} + 1\right\}$$

Since $L(A) = \begin{cases} 1, & Max\{x_i\} - 1 < A < \min\{x_i\} + 1 \\ 0, & otherwise \end{cases}$

If we want to select the one that is *unbiased* we select the middle-point of the interval:

$$\hat{A}_{ML} = \frac{\left(Max\{x_i\} - 1\right) + \left(\min\{x_i\} + 1\right)}{2} = \frac{\min\{x_i\} + Max\{x_i\}}{2}$$

To calculate the bias for *N* generic and prove that this estimator is unbiased is more difficult and **not requested here**.